

Iridium

Iridium is the Slocum's satellite link and the **primary way you talk to a glider once it is deployed** and out of Freewave range. The glider carries an **Iridium 9522b** bidirectional satellite modem on the lower electronics tray; a low-noise-amplifier (LNA) switch board lets the modem **share its quad-helix antenna with the GPS** (Iridium ~1626 MHz, GPS 1575 MHz), so the glider can fix its position and then call home through the same antenna in the tail.

Where **Freewave** is fast, free, and line-of-sight, Iridium is **slow and metered but global**: when the glider surfaces anywhere in the world with a clear view of the sky, it can reach your Dockserver.

Source

Paraphrased from the *Slocum G3 Glider Operators Manual* (Rev. 1, "Iridium Satellite Telemetry", "Glider Communications", and Appendices G and H), the UG2 community Slack, and the Teledyne Webb Research user forum. This is a condensed field reference — always defer to the official Teledyne documentation for your specific glider, and contact glidersupport@teledyne.com before changing SIM, phone-number, or Argos configuration.

How the call reaches shore: RUDICS vs. dial-up (PSTN)

When the glider dials in, energy leaves the tail antenna, **bounces satellite-to-satellite** across the Iridium constellation, and comes down at a **ground station**. What happens *after* the ground station is the difference between the two transports: